A Project Report on

REAL -WORLD PEDAL ENERGY HARVESTING

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ABSTRACT

The increasing demand for sustainable and renewable energy sources has encouraged the development of innovative energy-harvesting technologies. This project presents a real-world pedal energy harvesting system that converts mechanical energy generated from human footsteps into electrical energy using piezoelectric floor tiles. The system is designed to capture and store energy produced in high-footfall areas such as railway stations, shopping malls, airports, educational institutions, and public walkways.

Experimental data, including weight, step location, voltage, current, and generated power, were collected from the piezoelectric setup. Machine learning techniques were then applied to analyze the collected data and predict power generation under varying conditions. Several regression models, including Linear Regression, Decision Tree Regression, Random Forest Regression, and Gradient Boosting Regression, were trained and evaluated using performance metrics such as R^2 Score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

The results demonstrated that the Gradient Boosting Regression model achieved the highest prediction accuracy, with an R^2 score of approximately 0.96, indicating strong predictive capability. Correlation analysis further revealed that voltage and power output have a strong positive relationship, while step location significantly influences energy generation. The findings highlight the potential of combining piezoelectric energy harvesting with machine learning to optimize power prediction and improve the efficiency of smart energy systems.

This project contributes to the advancement of green energy solutions by demonstrating an intelligent approach to harvesting and predicting renewable energy generated from everyday human activities.

INTRODUCTION

The rapid growth in global energy consumption and the increasing concerns about environmental sustainability have created a strong need for alternative and renewable energy sources. Traditional energy generation methods largely depend on fossil fuels, which contribute to environmental pollution and the depletion of natural resources. As a result, researchers are exploring innovative techniques to harvest energy from everyday human activities.

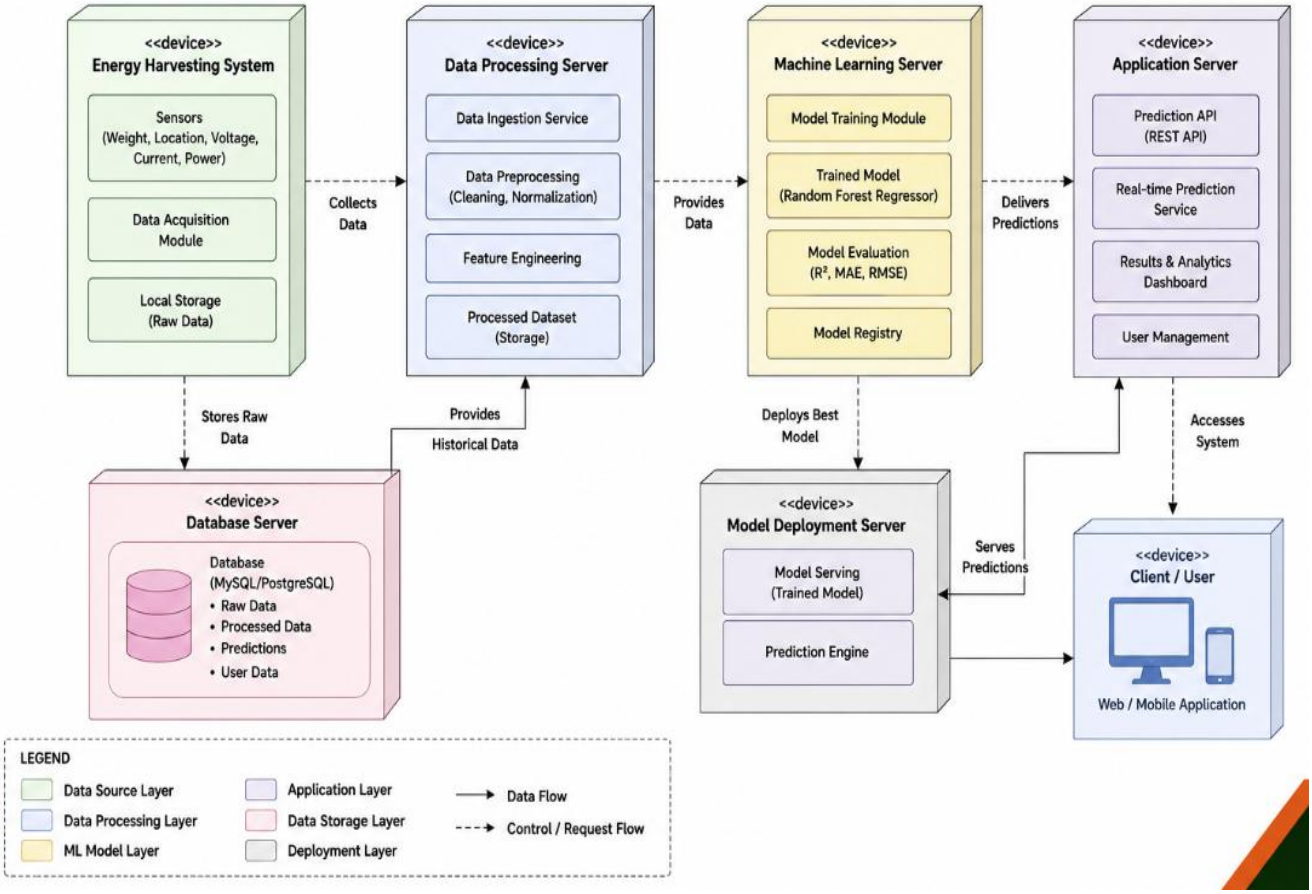
One promising approach is **piezoelectric energy harvesting**, which converts mechanical stress into electrical energy. Piezoelectric materials generate an electric charge when subjected to pressure or vibration. By embedding piezoelectric sensors beneath floor tiles, electrical energy can be produced from human footsteps. This technology is particularly useful in crowded locations such as railway stations, shopping malls, airports, schools, and public walkways, where large numbers of people generate continuous mechanical pressure.

While piezoelectric systems can effectively generate energy, the amount of power produced depends on several factors, including the weight of the individual, step location, walking pattern, and the characteristics of the piezoelectric material. Predicting the generated power accurately is essential for designing efficient energy-harvesting systems and optimizing their performance.

In this project, a real-world pedal energy harvesting system is developed using piezoelectric floor tiles to capture energy from footsteps. Experimental data such as weight, step location, voltage, current, and power output are collected and analyzed. Machine learning algorithms are then employed to predict power generation based on the collected parameters. Various regression models, including Linear Regression, Decision Tree Regression, Random Forest Regression, and Gradient Boosting Regression, are implemented and compared to identify the most accurate prediction model.

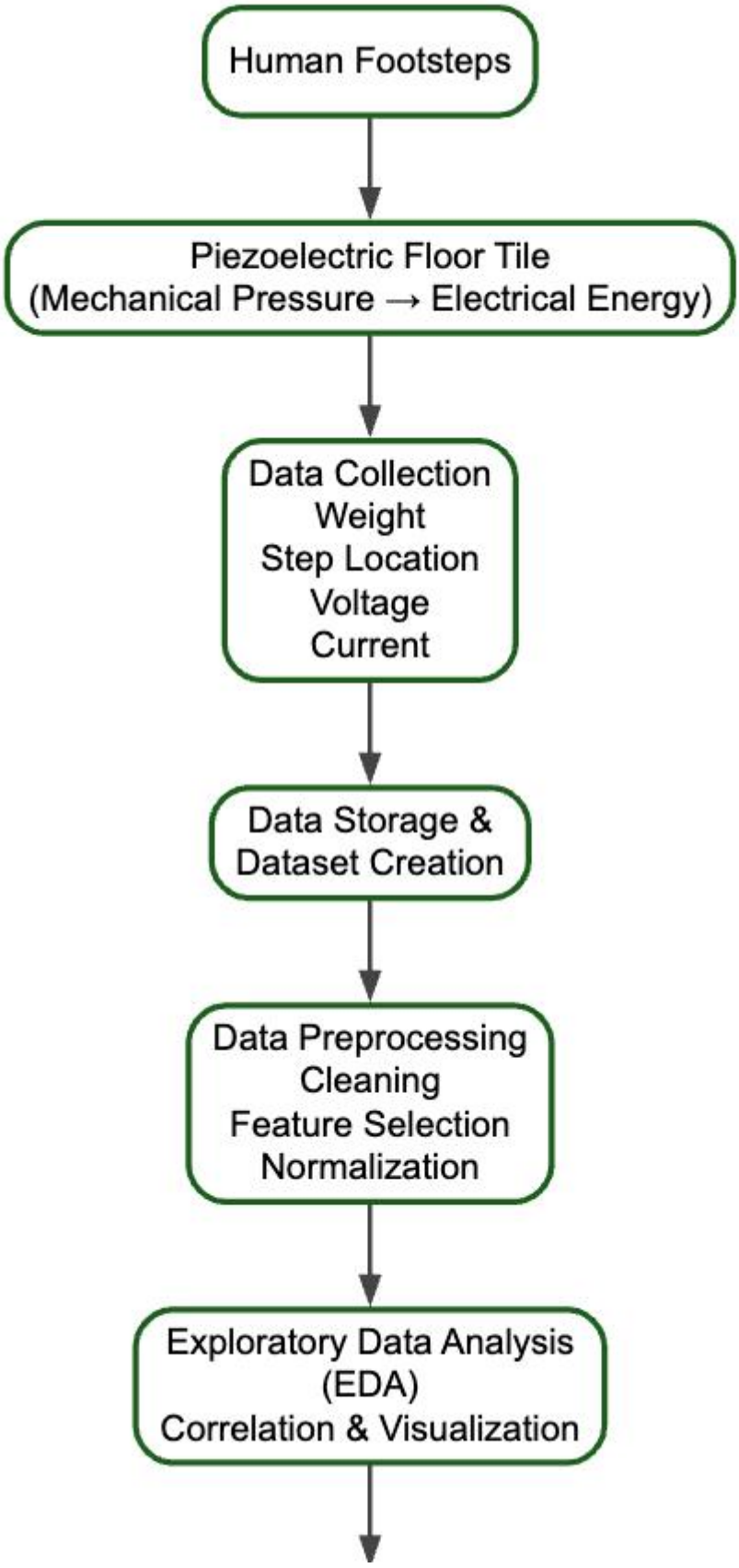
The integration of machine learning with piezoelectric energy harvesting provides an intelligent and data-driven approach for optimizing renewable energy generation. This project demonstrates how advanced predictive analytics can enhance the efficiency and reliability of sustainable energy systems while promoting environmentally friendly technologies for future smart cities and energy-aware infrastructures.

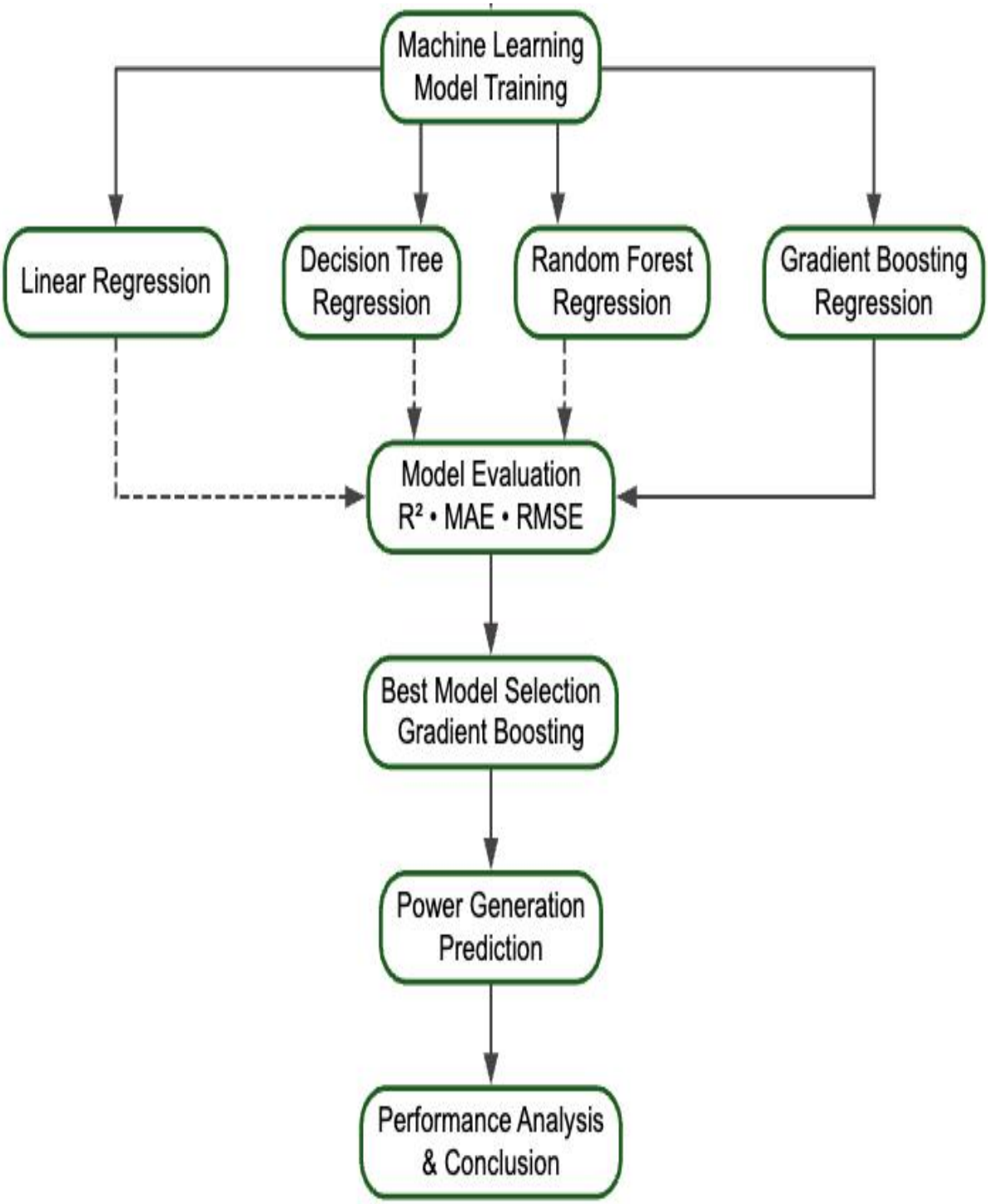
SYSTEM ARCHITECTURE



The workflow begins with **human footsteps** applying pressure on the piezoelectric tile, which converts mechanical energy into electrical energy. The generated voltage and current are collected and stored to create the dataset. The collected data is then preprocessed and enhanced through feature engineering before being used for model training. Multiple machine learning algorithms are trained and evaluated to identify the best-performing model. Finally, the selected model is used to predict power generation, and its performance is analyzed using standard evaluation metrics.

PROJECT WORKFLOW




Machine Learning Model Training
Linear Regression
Decision Tree Regression
Random Forest Regression
Gradient Boosting Regression
Model Evaluation
R² • MAE • RMSE
Best Model Selection
Gradient Boosting
Power Generation Prediction
Performance Analysis & Conclusion

SOFTWARE REQUIREMENTS

Software/Tool	Purpose
Python 3	Programming language used for data analysis and machine learning implementation
Jupyter Notebook / Google Colab	Development and execution environment
Pandas	Data collection, cleaning, and manipulation
NumPy	Numerical computations and array operations
Matplotlib	Data visualization and plotting graphs
Seaborn	Statistical data visualization and correlation analysis
Scikit-learn (Sklearn)	Machine learning model development and evaluation

DATASET OVERVIEW AND FEATURE ENGINEERING

Dataset Overview:

The dataset used in this project was created by collecting real-time measurements from a piezoelectric footstep energy harvesting system. The purpose of the dataset is to analyze the relationship between footstep characteristics and the electrical power generated by the piezoelectric sensors. Each record in the dataset represents a single footstep event and contains the corresponding electrical output parameters.

The collected data includes information such as the weight applied on the tile, the location of the step, the generated voltage, the generated current, and the resulting power output. These parameters are used to train machine learning models for accurate power generation prediction.

Dataset Features

Feature Name	**Description**	**Type**
Weight (kg)	Weight applied on the piezoelectric tile	Numerical
Step Location	Position of the footstep (Center, Edge, Corner)	Categorical
Voltage (V)	Voltage generated by the piezoelectric sensor	Numerical
Current (A)	Current generated by the piezoelectric sensor	Numerical
Power (W)	Electrical power generated (Target Variable)	Numerical

Feature Engineering:

Feature engineering is a crucial step in machine learning that improves model performance by transforming raw data into meaningful features.
The following feature engineering techniques were applied in this project:

1. Data Cleaning

- Checked for missing values and removed inconsistencies.
- Verified sensor readings to eliminate erroneous measurements.
- Ensured uniform data formatting.

2. Encoding Categorical Variables

The **Step Location** feature is categorical and cannot be directly processed by machine learning algorithms. Therefore, it is converted into numerical form using encoding techniques.

Example:

Step Location	**Encoded Value**
Center	0
Edge	1
Corner	2

3. Feature Selection

The most relevant features influencing power generation were selected:

- Weight
- Step Location
- Voltage
- Current

These features were chosen based on their direct impact on electrical power output.

4. Correlation Analysis

A correlation matrix was generated to identify relationships among variables. The analysis revealed:

- Strong positive correlation between Voltage and Power.
- Moderate correlation between Current and Power.
- Significant influence of Step Location on generated energy.

5. Target Variable Definition

The output variable for prediction is:

$$Power = Voltage \times Current$$

The machine learning models are trained to predict power generation based on the input features.

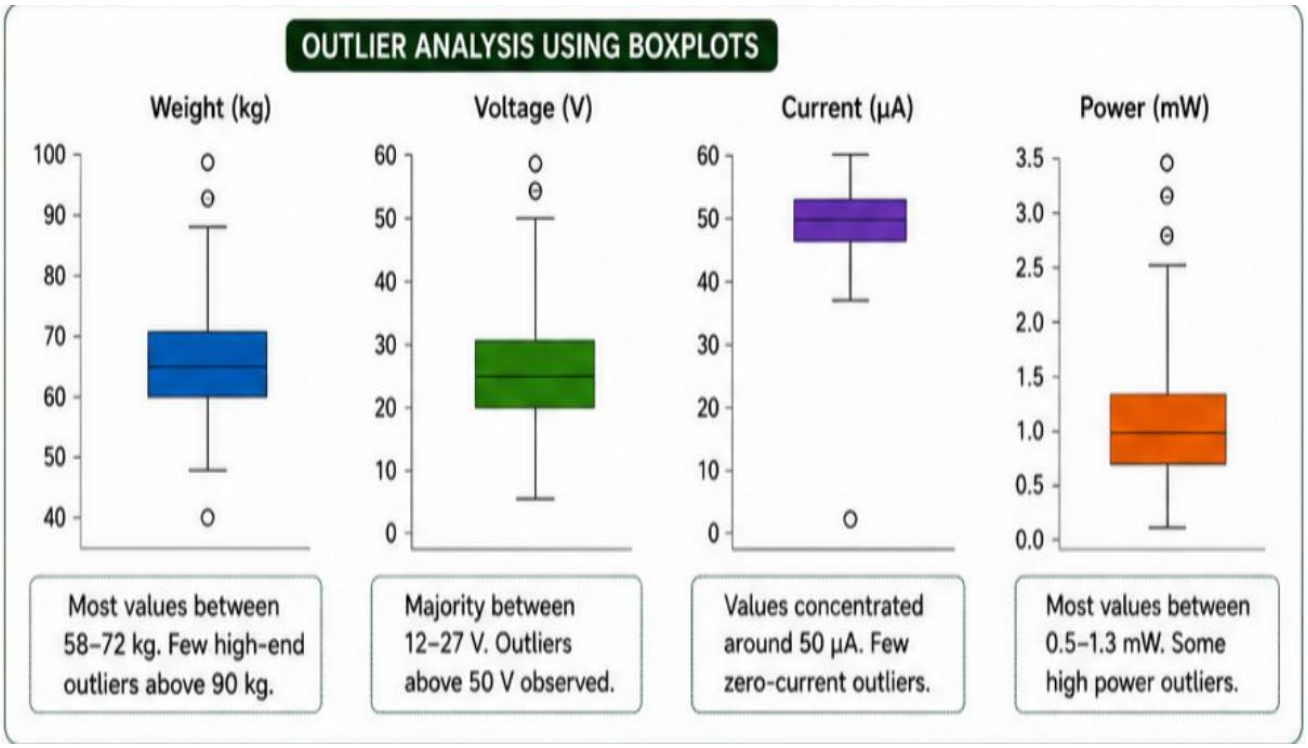
6. Data Splitting

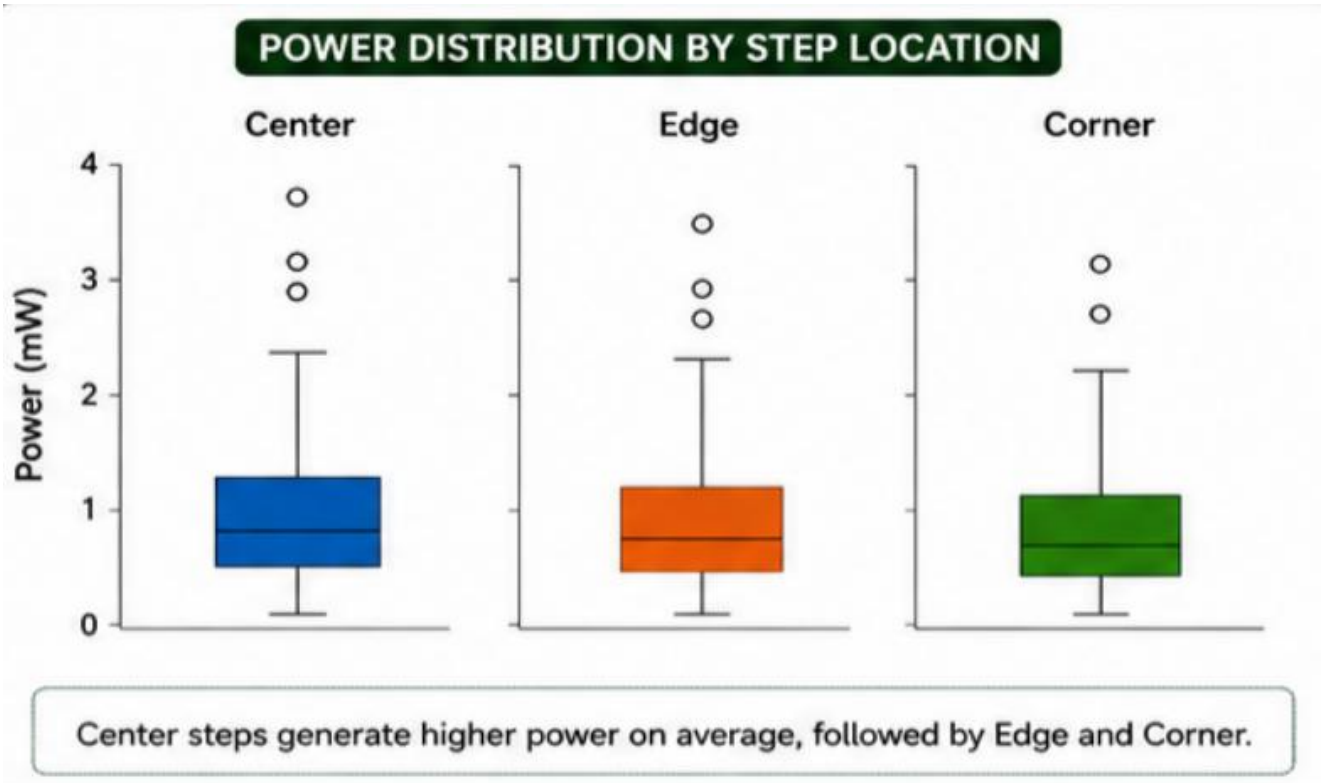
The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

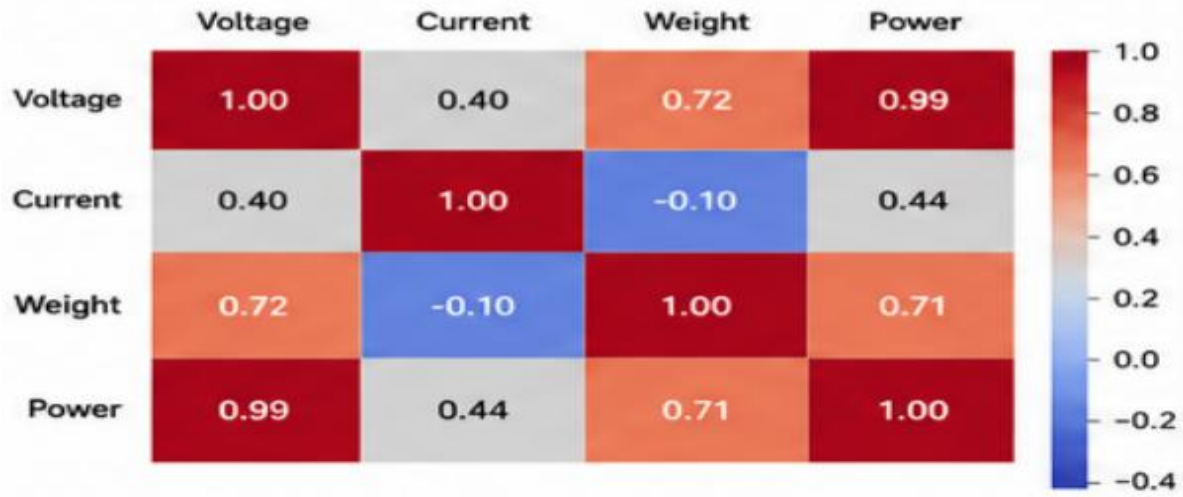
This ensures that the models are evaluated on unseen data and can generalize effectively.

DATA ANALYSIS





CORRELATION MATRIX (ORIGINAL FEATURES)



Voltage has the strongest correlation with Power (0.99), followed by Weight (0.71). Current shows weak correlation.

CORRELATION MATRIX (AFTER FEATURE ENGINEERING)

	Voltage	Current	Weight	Power	Pressure Index	Current Utilization	Voltage Stability	Efficiency Score
Voltage	1.00	0.40	0.72	0.99	0.55	0.40	-0.03	0.19
Current	0.40	1.00	-0.10	0.44	-0.16	1.00	0.34	0.76
Weight	0.72	-0.10	1.00	0.71	0.67	-0.10	-0.20	-0.19
Power	0.99	0.44	0.71	1.00	0.53	0.44	-0.04	0.20
Pressure Index	0.55	-0.16	0.67	0.53	1.00	-0.16	-0.19	-0.21
Current Utilization	0.40	1.00	-0.10	0.44	-0.16	1.00	0.34	0.76
Voltage Stability	-0.03	0.34	-0.20	-0.04	-0.19	0.34	1.00	0.87
Efficiency Score	0.19	0.76	-0.19	0.20	-0.21	0.76	0.87	1.00

1.0 0.5 0.0 −0.5 −1.0

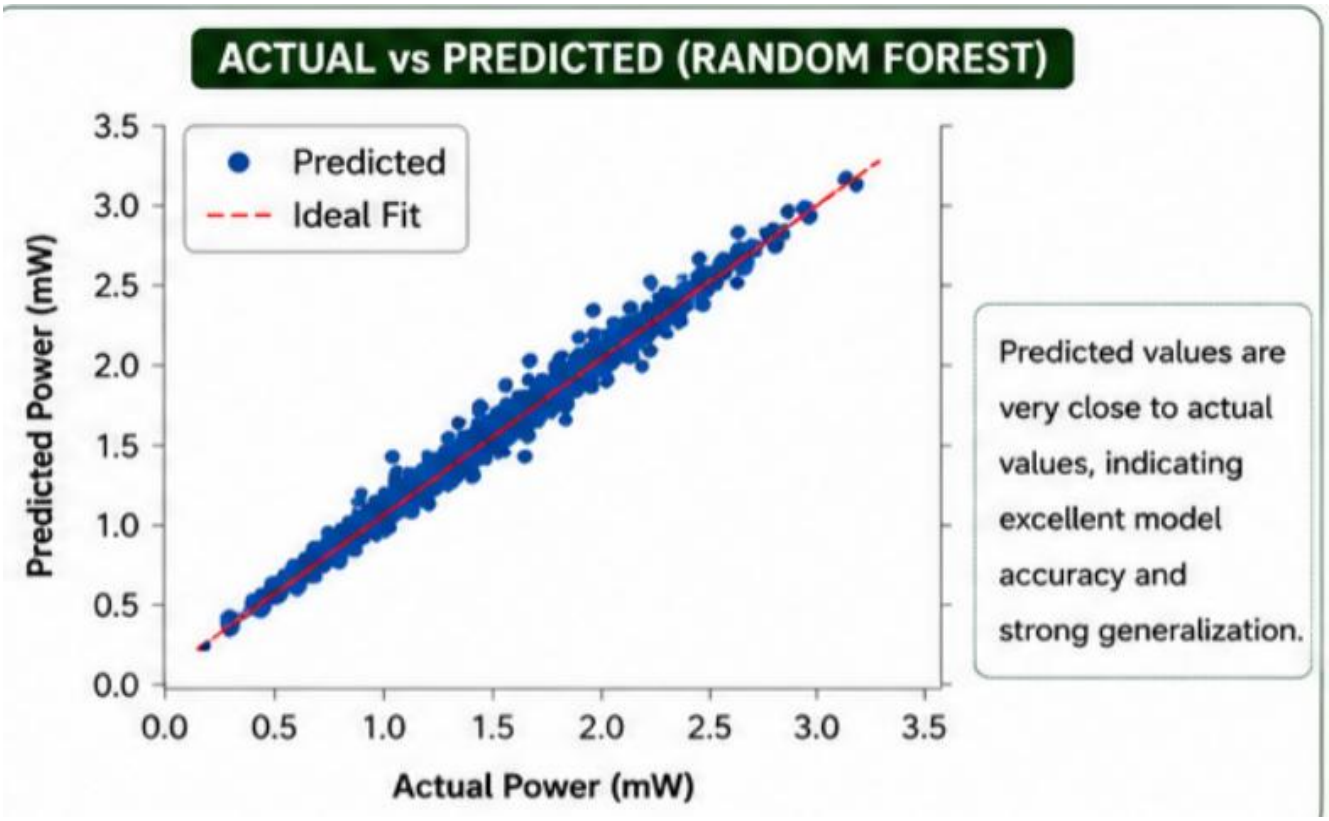
Engineered features show meaningful relationships with Power and each other. Efficiency Score is strongly correlated with Voltage Stability (0.87) and Current Utilization (0.76).

MODEL PERFORMANCE

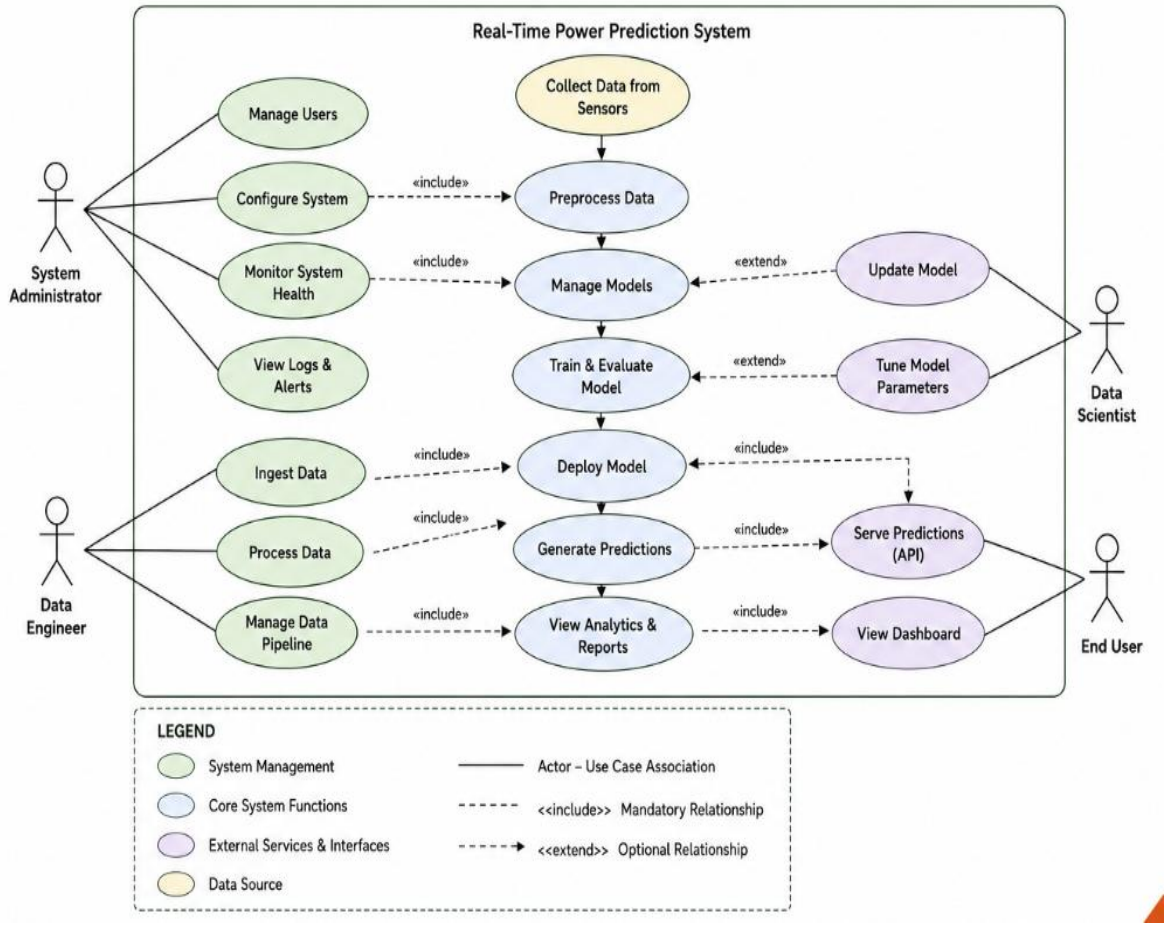
MODEL COMPARISON SUMMARY

Rank	Model	R^2 Score (↑)	MAE (↓)	RMSE (↓)
1	Random Forest Regressor	0.9685	0.0399	0.0990
2	Gradient Boosting Regressor	0.9542	0.0700	0.1194
3	Decision Tree Regressor	0.9405	0.0478	0.1362
4	Linear Regression	0.7839	0.2029	0.2594

Higher R^2 Score and lower MAE/RMSE indicate better performance.



USE CASE DIAGRAM



The use case diagram represents the functional interaction between the user and the proposed power prediction system. The user activates the piezoelectric tile by applying foot pressure, generating electrical data that is collected and processed. After preprocessing and feature engineering, multiple machine learning models are trained to predict power generation. The system evaluates the prediction results and provides accurate performance metrics for analysis and decision-making.

FUTURE SCOPE

1. IoT-Based Real-Time Monitoring

Integrate IoT devices to enable real-time monitoring of voltage, current, and power generation. This will allow remote tracking and efficient system management.

2. Large-Scale Smart City Deployment

Deploy the piezoelectric energy harvesting system in high-footfall locations such as railway stations, airports, shopping malls, and smart city infrastructure to maximize energy generation.

3. Advanced Machine Learning Models

Explore advanced algorithms such as XGBoost, LightGBM, and deep learning models to further improve the accuracy and robustness of power prediction.

4. Hybrid Renewable Energy Systems

Combine piezoelectric energy harvesting with other renewable sources such as solar or wind energy to develop a more reliable and sustainable hybrid energy solution.

5. Mobile Dashboard and Cloud Integration

Develop a cloud-based dashboard or mobile application for real-time visualization, analytics, and prediction of power generation, enabling efficient monitoring and decision-making.

THANK YOU